The documentation and process conversion measures necessary to comply with this revision shall be completed by 3 Mar 94.

INCH-POUND

MIL-S-19500/349C
3 December 1993
SUPERSEDING
MIL-S-19500/349B
26 May 1987

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING
TYPES 2N3506 AND 2N3507, JANTX, JANTXV AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, switching transistors. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings.

P_T 1/ T_A = +25°C	P_T 2/ T_C = +25°C	V_{CBO}		V_{CEO}		V_{EBO}	I_C	T_J and T_{STG}
		2N3506	2N3507	2N3506	2N3507			
W	W	V dc	V dc	V dc	V dc	V dc	A dc	°C
1.0	5.0	60	80	40	50	5.0	3.0	-65 to +200

1/ Derate linearly 5.71 mW/°C above T_A = +25°C.
2/ Derate linearly 28.6 mW/°C above T_C = +25°C.

1.4 Primary electrical characteristics.

	h_{FE2} 1/		h_{FE4} 1/		$V_{CE(sat)}$	$\lvert h_{fe} \rvert$	Cobo	t_{on}	t_{off}
Limits	V_{CE} = 2.0 V dc I_C = 1.5 A dc		V_{CE} = 5.0 V dc I_C = 3.0 A dc		I_C = 1.5 A dc I_B = 150 mA dc	f = 20 MHz V_{CE} = 5 V dc I_C = 100 mA dc	V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	I_C = 1.5 A dc I_B = 150 mA dc	
	2N3506	2N3507	2N3506	2N3507	V dc		pF	ns	ns
Min	40	30	25	20		3			
Max	200	150			1.0	15	40	45	90

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270 by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 8.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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4.3 Screening. Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	h_{FE2}, I_{CEX1}	N/A
11	I_{CEX1}; h_{FE2}; ΔI_{CEX1} = 100% of initial value or 200 nA dc, whichever is greater; Δh_{FE2} = 15% of initial value	I_{CEX1} and h_{FE2}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CEX1} = 100% of initial value or 200 nA dc, whichever is greater; Δh_{FE2} = 15% of initial value.	Subgroup 2 of table I herein; ΔI_{CEX1} = 100% of initial value or 200 nA dc, whichever is greater; Δh_{FE2} = 15% of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = +25°C ± 3°C; V_{CB} = 30 V dc; P_T = 1.0 W. JANS devices - - V_{CB} = 10 V dc.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
B3	2037	Test condition A
B4	1037	V_{CB} = 10 V dc; T_A = room ambient as defined in the general requirements of (see 4.5) of MIL-STD-750; P_T = 1.0 W; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on the devices shall be permitted.
B5	1027	V_{CB} = 10 V dc; T_A = +125°C ± 25°C for 96 hours, P_T = 1.0 W at T_A = +100°C or adjusted as required according to the chosen T_A to give an average T_J = +275°C.
B6	3131	See 4.5.2

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4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B3	1027	V_{CB} = 10 V dc; P_T = 1.0 W at T_A = room ambient as defined in the general requirements of (see 4.5) of MIL-STD-750. No heat sink or forced-air cooling on the devices shall be permitted.
B4	2037	Test condition A

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Condition
C2	2036	Test condition E
C6	1026	V_{CB} = 10 V dc; P_T = 1.0 W at T_A = room ambient as defined in the general requirements of (see 4.5) of MIL-STD-750. No heat sink or forced-air cooling on the devices shall be permitted.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Minimum collector current magnitude during power application shall be 160 mA dc for $R_{\theta JC}$ and 32 mA dc for $R_{\theta JA}$.

b. Collector to base voltage magnitude shall be 10 V dc.

c. Reference temperature measuring point shall be the case for $R_{\theta JC}$ and ambient air for $R_{\theta JA}$.

d. Reference point temperature shall be selected with 25° ≤ T_R ≤ 35°C and recorded before test is started.

e. Mounting arrangement shall be with heat sink to case for $R_{\theta JC}$ and without heat sink for $R_{\theta JA}$.

f. Maximum limits shall be $R_{\theta JC}$ = 35°C/W and $R_{\theta JA}$ = 175°C/W.

NOTE: Unless otherwise specified by the acquiring activity, the choice for measuring $R_{\theta JC}$ or $R_{\theta JA}$ will satisfy the requirements of this test.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage collector to base	3001	Bias condition D, I_C = 100 µA dc	$V_{(BR)CBO}$			
2N3506				60		V dc
2N3507				80		V dc
Breakdown voltage emitter to base	3026	Bias condition D, I_E = 10 µA dc	$V_{(BR)EBO}$	5		V dc
Breakdown voltage collector to emitter	3011	Bias condition D, I_C = 10 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N3506				40		V dc
2N3507				50		V dc
Collector to emitter cutoff current	3041	Bias condition A, V_{EB} = 4 V dc	I_{CEX1}		1	µA dc
2N3506		V_{CE} = 40 V dc				
2N3507		V_{CE} = 60 V dc				
Forward-current transfer ratio	3076	V_{CE} = 1 V dc, I_C = 500 mA dc, pulsed (see 4.5.1)	h_{FE1}			
2N3506				50	250	
2N3507				35	175	
Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 1.5 A dc, pulsed (see 4.5.1)	h_{FE2}			
2N3506				40	200	
2N3507				30	150	
Forward-current transfer ratio	3076	V_{CE} = 3 V dc, I_C = 2.5 A dc, pulsed (see 4.5.1)	h_{FE3}			
2N3506				30		
2N3507				25		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 3.0 A dc, pulsed (see 4.5.1)	h_{FE4}			
2N3506				25		
2N3507				20		

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Saturation voltage and resistance	3071	I_C = 500 mA dc, I_B = 50 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.5	V dc
Saturation voltage and resistance	3071	I_C = 1.5 mA dc, I_B = 150 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.0	V dc
Saturation voltage and resistance	3071	I_C = 2.5 A dc, I_B = 250 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)3}$		1.5	V dc
Base to emitter saturated voltage	3066	Test condition A, I_C = 500 mA dc, I_B = 50 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.0	V dc
Base to emitter saturated voltage	3066	Test condition A, I_C = 1.5 A dc, I_B = 150 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$	0.9	1.4	V dc
Base to emitter saturated voltage	3066	Test condition A, I_C = 2.5 A dc, I_B = 250 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)3}$		2.0	V dc
Subgroup 3						
High-temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A, V_{EB} = 4 V dc	I_{CEX2}		1.5	mA dc
2N3506		V_{CE} = 40 V dc				
2N3507		V_{CE} = 60 V dc				
Low-temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc, I_C = 500 mA dc, pulsed (see 4.5.1)	h_{FE5}			
2N3506				25		
2N3507				17		
Subgroup 4						
Small-signal short- circuit forward-current transfer ratio	3206	V_{CE} = 5 V dc, I_C = 100 mA dc, f = 20 MHz	$\|h_{fe}\|$	3	15	

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Open circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		40	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 3 V dc, I_C = 0, 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		300	pF
Pulse response:						
Delay time	3251	Test condition A, I_C = 1.5 A dc, I_{B1} = 150 mA dc, (see figure 1)	t_d		15	ns
Rise time	3251	Test condition A, I_C = 1.5 A dc, I_{B1} = 150 mA dc, (see figure 1)	t_r		30	ns
Storage time	3251	Test condition A, I_C = 1.5 A dc, I_{B1} = I_{B2} = 150 mA dc, (see figure 2)	t_s		55	ns
Fall time	3251	Test condition A, I_C = 1.5 A dc, I_{B1} = I_{B2} = 150 mA dc, (see figure 2)	t_f		35	ns

1/ For sampling plan see MIL-S-19500.

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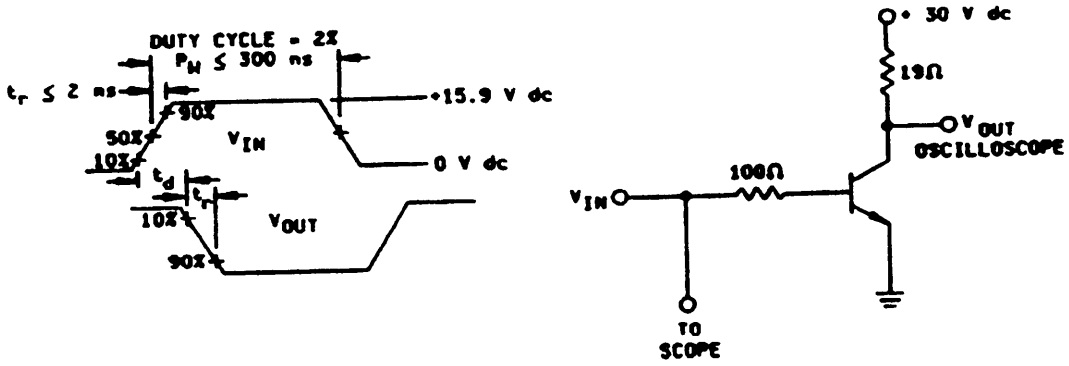


FIGURE 1. Equivalent circuit for measuring delay and rise times.

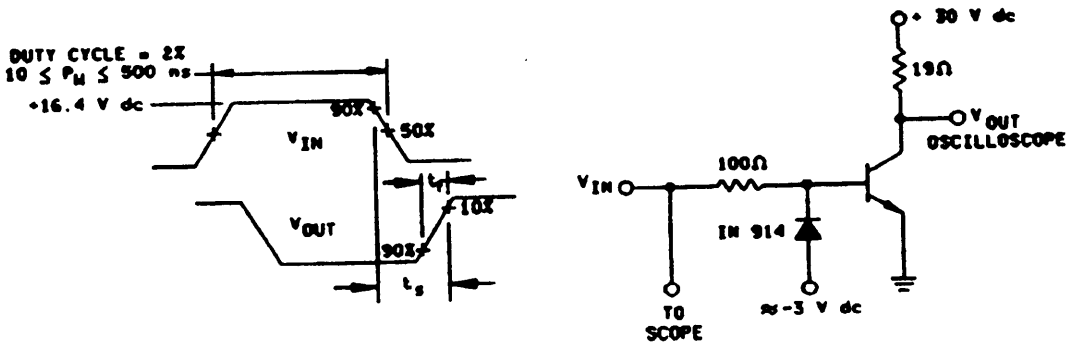


FIGURE 2. Equivalent circuit for measuring storage and fall times.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Lead finish as specified (see 3.3.1).

b. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, AV, MI
Navy - AS, CG, MC
Air Force - 13, 19, 85
DLA - ES
NASA - NA

Preparing activity:
Navy - EC

Agent:
DLA - ES

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